



Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Major Examination

AY 25-26

Semester II

Full Marks 45

Duration: 180 Mins

A. Objective Questions

1. Please collect the separate copy for the objective questions. Write your answers on it. Submit it along with the main answer sheet. (9)

B. Subjective Type

2. Solve the following linear system of first-order homogeneous ordinary differential equations:

$$\begin{cases} x' = 6x - 8y \\ y' = 2x - 2y. \end{cases}$$

(3)

3. Determine at least one solution to

$$x^2 y'' + xy' - (4+x)y = 0, \quad x > 0,$$

by determining n -th term. Also, write down the form of the second solution, which is linearly independent. (3)

4. Determine the general solution to

$$x^3 y''' - 3x^2 y'' + 7xy' - 8y = 0, \quad x > 0.$$

(3)

5. Determine the general solution to the differential equation

$$y'' + y = \sec x + 4e^x, \quad |x| < \frac{\pi}{2}.$$

(3)

6. Determine the values of r and s such that $I(x, y) = x^r y^s$ is an integrating factor for the differential equation

$$2y(y + 2x^2)dx + x(4y + 3x^2)dy = 0.$$

Then, also solve the above ODE. (3)

7. Prove or disprove: A linear combination of a set of eigenvectors of a matrix A is again an eigenvector. (3)



8. Let L be a line segment joining (x_1, y_1) and (x_2, y_2) and $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. Derive the image of L under T .

(3)

9. Let $W = \{p(B) : p \text{ is a polynomial with real coefficients}\}$, where $B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Find the dimension of the vector space W .

(3)

10. If V_1 is the xy -plane, V_2 is the z -axis, and V_3 is the line in $\mathbb{R}^3$ passing through $(0, 0, 0)$ and $(1, 1, 1)$, then can $\mathbb{R}^3$ be written as a direct sum of V_1, V_2 and V_3 ?

(3)

11. Give an example of a linear transformation which possesses all real numbers as its eigenvalues. Talk about its eigenvectors also.

(3)

12. Given real numbers $a_0, a_1, \dots, a_k$, let V be the set of all solutions $x \in C^k[a, b]$ of the differential equation

$$a_0 \frac{d^k x}{dt^k} + a_1 \frac{d^{k-1} x}{dt^{k-1}} + \dots + a_k x = 0.$$

Is V a vector space over $\mathbb{R}$? If yes, find $\dim(V)$?

(3)

13. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear map having eigenvalues 2, 3, 4. Justify whether true or false:

- there exists infinitely many $x \in \mathbb{R}^3$ such that $Tx = (a_1, a_2, a_3)^T$, where $a_i \in \mathbb{R}$ for $i = 1, 2, 3$.

(3)



Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Minor Examination

RCH
BARR.2H1

AY 25-26

Semester II

Full Marks 25

Duration: 120 Mins

A. Objective Questions

1. A separate copy is attached.

The answers of the objective questions should be written on the question paper and needs to be submitted with the answer sheet.

(5)

B. Subjective Type

2. Determine values of the constant k for which the given set of vectors is linearly dependent in $\{(1, 1, 0, -1), (1, k, 1, 1), (2, 1, k, 1), (-1, 1, 1, k)\}$.

(2)

3. If A is an $m \times n$ matrix and b is a nonzero element of $\mathbb{R}^m$, do the solutions to system $Ax = b$ form a subspace of $\mathbb{R}^n$? Why or why not?

(2)

4. Let V be an inner product space with the inner product $\langle \cdot, \cdot \rangle$. Then prove that

$$\langle x + y, x - y \rangle < 0 \text{ if and only if } \|x\| < \|y\|.$$

Can you comment on anything extra if we have the equality instead of the strict inequality? (1.5+0.5)

5. Check whether the map $\langle \cdot, \cdot \rangle$ defined on $V \times V$ as

$$\langle A, B \rangle = \text{trace}(B^T A)$$

is an inner product or not. Where $V = M_{3 \times 2}(\mathbb{R})$ (set of all 3×2 matrices).

(2)

6. Let $V = P_2$. Also suppose $\mathcal{B} = \{1, 1+x, 1+x+x^2\}$ and $\mathcal{C} = \{2+x+x^2, x+x^2, x\}$. Find the change-of-basis matrices from $\mathcal{B}$ to $\mathcal{C}$, and use it to calculate the change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$.

(1+1)

7. Let $T : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ be a linear transformation defined by

$$T(A) = BA - AB,$$

where

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

- (a) Find the matrix representation of T with respect to the standard basis.

- (b) Find the null space $\ker(T)$ and verify Rank-Nullity Theorem.

(1+1)

8. Let $T : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ be a linear transformation defined by

$$T(A) = A + A^T.$$

- (a) Find the null space $\ker(T)$ and interpret the null space geometrically.
- (b) Show that any matrix $A \in M_{2 \times 2}(\mathbb{R})$ can be written in the form

$$A = S + K,$$

where $S \in \text{Range}(T)$ and $K \in \ker(T)$.

(1+1)

9. Let $T : P_3 \rightarrow P_4$ be a linear transformation defined by

$$T(p(x)) = \int_0^x p(t) dt.$$

- (a) Find a basis for the range space $\text{Range}(T)$.
- (b) Find the null space $\ker(T)$.

(1+1)

10. Let $T : P_3 \rightarrow \mathbb{R}^3$ be defined by

$$T(p) = (p(0), p(1), p(2)).$$

Let $B = \{1, x, x^2, x^3\}$ be the standard basis of P_3 .

- (a) Interpret the null space in terms of the roots of polynomials.
- (b) Find all polynomials $p(x) \in P_3$ such that

$$T(p) = (1, 0, 1).$$

(1+1)

11. Let $T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}$ be defined by

$$T(A) = \text{trace}(A).$$

- (a) Find a basis for $\ker(T)$.

- (b) Find all matrices A such that

$$T(A) = 5.$$

(1+1)

Best Wishes